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TRANSFER OF RESISTANCE TO THE STREAK VIRUS INTO EAST AFRICAN MAIZE

By H. H. Storey and Audrie K. Howland, E.A.A.F.R.O., Kikuyu, Kenya

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Streak disease is caused by infection with a virus that is prevalent in maize and some grasses in many parts of Africa. It has been frequently described (e.g. Storey, 1925; McClean, 1947); typically it results in leaves developing with a nearly uniform pattern of chlorotic and green areas; in its most severe form some half or more of the total leaf area may be chlorotic, and in this form the disease causes a marked reduction in the growth and productivity of the plant. Because the virus is transmissible only by a specific insect vector of the genus *Cicadulina* (Storey, loc. cit.) and by no other known means, prevalence of the disease in the field is dependent not only on the presence of the virus but also on the frequency with which the vector occurs. In consequence, outbreaks of streak disease may vary from season to season and from place to place. Nevertheless, in aggregate the disease can cause large losses in productivity of maize; and a search for resistant maize types is justified.

Glasshouse studies of streak resistance in maize have been prosecuted over a number of years at E.A.A.F.R.O. That this resistance existed had been recognized in the early thirties in South Africa, but owing to difficulties inherent in studies of this problem under field conditions, research workers there had been unable to arrive at any hypothesis for the manner in which resistance was inherited (Rose 1938; Gorter, 1959). The purpose of the work at E.A.A.F.R.O. was to see whether a genetic hypothesis would emerge from studies carried out under the more exact, more reliable and more rapid conditions provided by glasshouses, with which E.A.A.F.R.O. was well equipped.

The South African work, deriving primarily from the initial discovery at Barberton of resistance in a variety entitled "Peruvian Yellow", had been developed to a point where satisfactory control of streak could be claimed (Gorter, 1959). First, the Peruvian Yellow had been crossed with Hickory King, and by selection resistant "P × H" lines had been developed that gave white seeds. In the later work, "P × H" had been crossed with a range of high-yielding but susceptible crosses from the departmental breeding programme. The resulting three-way crosses did not show a marked reduction in percentage infection but the severity of the disease was reported to be greatly reduced. Selections from this programme were issued to farmers in areas where the virus was prevalent.

Gorter (loc. cit) noted that breeding was handicapped by lack of genetic information on inheritance of resistance. The E.A.A.F.R.O. studies aimed at correcting this deficiency; and up to a point we believe they achieved this object. Inbred lines from imported "P × H" material appeared to carry a major gene for high resistance, the allelomorph of which gave high susceptibility. Neither was dominant and the hybrid was intermediate, showing phenotypes with moderately severe streak symptoms but suffering only limited ill-effects from infection by the virus. The policy adopted by South Africa depended upon this "tolerance" by the heterozygote (Gorter, loc. cit.).

Basic Studies

A full description of our glasshouse methods and of the conclusions from results obtained has been set out elsewhere (Story and Howland, 1967). For the present paper it is sufficient to report that young seedlings inoculated with the streak virus through the agency of the insect vector, *Cicadulina mbila*, could be classed in the following groups, described and illustrated in the paper mentioned above.

Resistant—Class "O" (no symptoms visible); Classes "A— —", "A—", "A+" (small chlorotic spots, circular or slightly elongated, in a series of increasing frequency in distribution on early formed leaves, but usually none on later leaves. No detectable ill-effect on the plants' growth at any stage).

Intermediate—Classes "B—" and "B+". (Size and frequency of chlorotic sports and streaks greater than in "A+", and this distribution usually maintained on each leaf formed to maturity. Ill-effects on plants' growth scarcely detectable by inspection.) In addition

evidence showed that a proportion of the plants classed "C—" could properly be included in this group (see below).

Susceptible—Normally classed "C+" (the most severe form of streak disease, chlorotic spots and streaks occupying some half of the total leaf area, but tending to a reduction in frequency in the latest leaves formed as the plants develop to maturity). In addition a class "C—" was recognized which approached "C+" in severity and was often not certainly separated from it by inspection; genetically "C—" was found to fall in part in the "Intermediate" and part in the "Susceptible" group.

Our breeding studies led to the conclusions that plants in the above "Resistant" group were generally homozygous for the resistant major gene, "intermediate" generally heterozygous, and "susceptibles" homozygous for the major allelomorph.

Materials

Resistant parents in the present programme were derived from second or third generation inbreds from the Peruvian × Hickory introductions: mainly used was a cross between two S_2 families of differing origin, both showing a narrow distribution in the resistant classes. This cross appeared to possess some hybrid vigour and in small-scale trials yielded better than other inbreds sampled.

Susceptible varieties constituting the recurrent parents were:—

AFRO No. 293—*Katumbili*. A race widely grown in Tanzania; samples used in the programme originated at Mwanza and Nachingwea.

AFRO No. 286—*Early Katumani Synthetic*. An issue from the breeding programme at Katumani, Kenya.

AFRO No. 290—*Muratha*. A race adapted to the Kenya highlands and widely grown in the lower areas of the Kiambu-Embu region. The sample used originated at the Embu Agricultural Station.

AFRO No. 289—*Githigu*. From the same region as No. 290, but adapted to higher altitudes; sample originated at Kiambu, Kenya.

AFRO No. 275—*Eldoret Yellow*. Adapted to the plateau region of Kenya. Sample originated in a breeding station at Eldoret.

AFRO No. 284—*Zapalote Chico*. An introduction to Muguga from Mexico. Adapted to low altitudes in Central America. (This line is believed to possess an additional advantage in carrying a major gene for resistance to the tropical rust, *Puccinia polysora*.)

Procedure

This followed a conventional backcrossing programme. An F1 generation was raised from each susceptible variety by pollinating with mixed pollen from each resistant line. Usually equal quantities of seeed from each hybrid plant were mixed to give an "F1 Bulk". Samples of each mixed bulk, varying between 20 and 200 in number of seeds, were then raised in glasshouses and tested by our normal procedure.

The susceptible parents on initial tests had all reacted in the "C" classes. The F1 hybrids were mostly in classes "B" and "C—", with a small "overflow" into "A" classes and into "C+". These results conformed with our conclusions that resistance as a whole was determined primarily by a major gene but that modifying factors led to some diversity in the family distributions.

By selecting from the bulk tests the plants that fell into the lowest classes "A" and "B—", we hoped to accumulate favourable minor genes; while, by bulking the progeny of similar crosses, we tended to maintain the genetic diversity possessed by the susceptible adapted parental lines.

From the F1 Bulk, selected plants were raised to maturity and pollinated with mixed pollen from a number of plants of the susceptible parental line, giving the first backcrossed generation, BC1. Again these were bulked and tested, and the most resistant plants selected for growing to maturity. This first backcross would be expected to be segregating in a ratio of 1:1 heterozygotes to susceptible homozygotes. By selection, we expected to disturb this balance in favour of the heterozygotes.

A second backcross generation was raised by a similar procedure, giving a selected BC2 Bulk for each of the breeding lines.

The next stage was to self-out pure breeding resistant lines. Samples of some 50-200 BC2 seeds were grown in the glasshouses, infected, and again the most resistant plants grown to maturity. These were self-pollinated and each ear retained shelled into a separate bag. Samples from each number were tested in the standard manner.

Because of the selection that had occurred in earlier generations, a proportion of the BC2 progenies were expected to be genotypes heterozygous for the resistance major gene; these genotypes should on selfing give families segregating one homozygous resistant in four (subject to some complications due to modifying factors).

In fact, the BC2/S1 generation was disappointing in that few families on test appeared to be uniformly resistant. Consequently, from the most resistant phenotypes second-selfed generations were raised to give the BC2/S2 generation. In this a number of families were obtained in which a preponderance of the resistant classes "O", "A— —", "A—" and "A+" occurred. In several, there tended to be some "overflow" into the "B" classes; but these aberrant phenotypes could be fairly attributed to the effect of modifying factors.

Data Obtained from the Breeding Programme

In Table I results are summarized in the form of total distributions observed in tests of the recurrent parental types and the five generations resulting from the procedure described above.

TABLE I—SUMMARY OF GENERATION TESTS OF SIX LINES

Susceptible Lines	Generation	Total Plants Tested	TOTAL 30-DAY DISTRIBUTION—PER CENT			
			"O"–"A"	"B"	"C—"	"C+"
293 (Katumbili)	P	12	0	0	33·3	66·7
	F1	24	0	16·7	79·2	4·2
	BC1	99	1·0	13·1	80·8	5·1
	BC2	91	2·2	17·6	65·9	14·3
	BC2/S1	59	33·9	32·2	16·9	16·9
	BC2/S2	279	98·9	0·7	0·4	0
286 (Early Katumani Synthetic)	P	12	0	0	41·7	58·3
	F1	22	18·0	22·7	54·5	4·5
	BC1	77	5·2	13·0	76·7	5·2
	BC2	115	0·9	40·0	52·2	7·0
	BC2/S1	142	36·6	31·0	17·6	14·8
	BC2/S2	239	90·4	9·6	0	0
290 (Muratha)	P	12	0	0	91·7	8·3
	F1	223	1·8	29·1	68·6	0·4
	BC1	102	2·9	17·6	69·6	9·8
	BC2	117	1·7	12·0	76·9	9·4
	BC2/S1	91	49·4	28·6	17·6	4·4
	BC2/S2	174	54·6	44·8	0·6	0

TABLE I—SUMMARY OF GENERATION TESTS OF SIX LINES—(*Contd.*)

Susceptible Lines	Generation	Total Plants Tested	TOTAL 30-DAY DISTRIBUTION—PER CENT			
			"O"–"A"	"B"	"C—"	"C+"
289 (Githigu)	P	11	0	0	54·5	45·5
	F1	88	2·3	30·7	65·9	1·1
	BC1	105	2·9	16·2	73·3	7·6
	BC2	122	2·5	9·8	77·9	9·8
	BC2/S1	53	30·2	34·0	20·7	15·1
	BC2/S2	159	33·3	38·4	21·4	6·9
275 (Eldoret Yellow)	P	12	0	0	16·7	83·3
	F1	24	8·3	29·2	62·5	0
	BC1	105	4·8	27·6	64·8	2·8
	BC2	91	7·7	47·2	40·7	4·4
	BC2/S1	136	27·2	41·9	18·4	12·5
	BC2/S2	332	49·7	43·4	6·9	0
284 (Zapalote Chico)	P	12	0	0	41·7	58·3
	F1	24	8·3	33·3	58·3	0
	BC1	102	2·0	7·8	70·6	19·6
	BC2	126	0·8	34·1	49·2	15·9
	BC2/S1	187	25·1	32·6	28·9	13·4
	BC2/S2	174	72·4	27·6	0	0

A study of Table I shows that our expectations were achieved: that is, the generations behaved generally in accordance with the hypothesis of resistance controlled primarily by a major gene but with considerable diversity in generation distributions due to modifying factors.

Almost throughout the table, there is a marked deficiency of the "C+" classes expected in families derived from crossing with a susceptible parent. In part we attribute this to observational errors in differentiating "C+" and "C—". (We have already referred to the doubts in eye-assessment of these two classes.) It will be noted that the chosen susceptible lines, inbred, themselves showed considerable differentiation within the "C" classes, from 17 per cent in No. 275 to 92 per cent in No. 290 falling in class "C—": in none of these was there any overflow into the lower classes. It is probable that many of the "C—" class phenotypes were in fact homozygous susceptibles. But, because in each generation selections were made of the most resistant phenotypes for further crossing, there had been a tendency throughout to reduce the numbers of homozygous susceptibles below those that would have been expected on simple genetic ratios.

On the other side of the table, deviations occurred that appeared again to be due to modifying factors. First, there was often a small spread into "A" classes in the F1, BC1 and BC2 generations, which should contain no homozygous resistant genotypes. In the selfed generation from BC2 there tended to be an unexpected extension from the "A" classes into "B"; but this repeated some of the findings from inbreeding believed pure resistant major genotypes in earlier studies.

The first selfed generation from the BC2, as already noted, contained few families with a high proportion of resistant phenotypes. The second selfs, however, gave families largely conforming with expectation: the majority of phenotypes falling into the resistant "O—A" classes but in some lines extending into the "B" class and higher. From this generation the most consistent families were selected, varying from three in No. 289 to eight in Nos. 293 and 286; the tests of these gave total distributions as in Table II.

The selected families were thus distributed solely in the "O—A" and the "B" classes. Overflow into the last class was insignificant in two lines (Nos. 293 and 286), but fairly considerable in the remaining four. There was no overflow into the "C" classes. The selected families in all lines therefore were effectively resistant.

From each of the six lines studied, the BC2/S2 families listed in Table II were bulked and were issued to the Kenya National Agricultural Research Station for further development in field-breeding by Mr. M. N. Harrison and his colleagues.

Summary

Glasshouse studies on breeding maize for resistance to the streak virus had led to the conclusion that control of this character was mainly by a major gene, but with some deviations attributed to minor genes acting as modifiers. On this basis, streak resistance was transferred to a group of maizes susceptible but otherwise adapted to East African environments; by crossing, backcrossing through two gerations and selfing selected phenotypes from the second backcross. Thirty-six families, deriving from six recurrent susceptible parental lines, were highly resistant, and were issued for further development to field-breeding stations.

Acknowledgements

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Table II—Sample Tests of BC2/S2 Families

Line	Families Selected		Total 30-Day Distributions of Selected Families—per cent			
	No. of Families	Total Plants	"O" to "A"	"B"	"C−"	"C+"
293	8	165	99·4	0·6	0	0
286	8	172	97·1	2·9	0	0
290	5	103	52·4	47·6	0	0
289	3	63	58·7	41·3	0	0
275	6	135	77·0	23·0	0	0
284	6	132	72·0	28·0	0	0